

Minchan Kim

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RESEARCH INTERESTS

Graph Representation Learning · Knowledge Graph Reasoning · Link Prediction · Graph-based Retrieval-Augmented Generation (GraphRAG) · Graph Condensation · LLM Reasoning & Interpretability

EDUCATION

Seoul National University <i>Ph.D. in Data Science</i> Advisor: Prof. Hyunwoo Park (DIAL Lab)	Mar. 2024 – Present Seoul, Republic of Korea
Seoul National University of Science & Technology <i>M.S. in Data Science</i> GPA: 4.45 / 4.5 Advisor: Prof. Hakyeon Lee (D&BI Lab)	Sept. 2022 – Feb. 2024 Seoul, Republic of Korea
Seoul National University of Science & Technology <i>B.S. in Computer Science and Engineering</i>	Mar. 2020 – Aug. 2022 Seoul, Republic of Korea
Seoul National University of Science & Technology <i>B.B.A. in Business Administration</i> GPA: 4.25 / 4.5 Rank: 1 / 30	Mar. 2017 – Aug. 2022 Seoul, Republic of Korea

PUBLICATIONS

Working Papers

- Minchan Kim**, Jungmin Hwang, and Hyunwoo Park. “What Gets Measured Gets Managed: Sign-aware Recommendation Needs Sign-aware Evaluation.” (Under review.)

Journal Articles

* These authors contributed equally.

- Minchan Kim**^{*}, Yanjie Zhao^{*}, Jaeseong Ju, Jaeun Seo, and Hyunwoo Park. “InteractiveKG: Transparent and User-Controllable Knowledge Graph Reasoning.” *Information Visualization*, 2026. (Special Issue from PacificVis 2026 Vis Meets AI Workshop.)
- Jeehong Kim^{*}, **Minchan Kim**^{*}, Youngseok Hwang, Sungho Bae, Deuk Jae Cho, Wonhee Lee, and Hyunwoo Park. “STGVAD: Spatio-Temporal Graph-Based Vessel Behavior Anomaly Detection.” *IEEE Access*, 2025.
- Minchan Kim** and Hakyeon Lee. “Treasure Hunting in the Talent Ocean: Automating Talent Acquisition for Competent Developers from GitHub.” *Annals of Data Science*, 2026.

Workshop Papers

- Jeehong Kim^{*}, Youngseok Hwang^{*}, **Minchan Kim**, Sungho Bae, and Hyunwoo Park. “Spatio-Temporal Graphs Beyond Grids: Benchmark for Maritime Anomaly Detection.” *NeurIPS 2025 Workshop on AI for Science*, San Diego, CA, USA, 2025.
- Jeehong Kim^{*}, **Minchan Kim**^{*}, Jaeseong Ju, Youngseok Hwang, Wonhee Lee, and Hyunwoo Park. “Adaptive Sparsified Graph Learning Framework for Vessel Behavior Anomalies.” *AAAI 2025 Workshop and Challenge on Anomaly Detection in Scientific Domains*, Philadelphia, PA, USA, 2025.

PATENTS

- “Developer Head-Hunting Candidate Automation System.” (Domestic) Application No. 0060095, 2023.

RESEARCH PROJECTS

Fraud Detection System (FDS) Modeling Optimization for Overseas Remittance Using AI MOIN	Jul. 2025 – Present
Vessel Behavior Anomaly Detection <i>Korea Research Institute of Ships & Ocean Engineering (KRISO)</i> Resulting publications: STGVAD (<i>IEEE Access</i>, 2025), with workshop papers at NeurIPS 2025 and AAAI 2025.	Oct. 2024 – Dec. 2025
Center for Optimizing Hyperscale AI Models and Platforms (ERC) <i>National Research Foundation of Korea (NRF) – Engineering Research Center</i> Resulting publication: InteractiveKG (<i>Information Visualization</i>, 2026).	Mar. 2024 – Present
Personalized Knowledge State Analysis & Question Recommendation with Knowledge Tracing Onuii	Jan. 2024 – Feb. 2024
ML-based Information Processing Analysis for B2B Business Environment Analysis <i>LG Electronics</i>	Sept. 2023 – Feb. 2024
ML-based Graph Learning & Link Prediction for Text-based Technology Trend Analysis <i>Hyundai Motor Company</i>	Jul. 2022 – May 2023
Nowcasting of Artificial Intelligence with Open Source Project Data <i>Ministry of Science and ICT</i> Resulting publication: “Treasure Hunting in the Talent Ocean” (<i>Annals of Data Science</i>, 2026).	Mar. 2022 – Feb. 2024
University Industry-Academia Research Capability Big Data Analysis <i>Seoul National University of Science & Technology</i>	Sept. 2021 – Dec. 2021

AWARDS & GRANTS

SNU Medical AI Scholarship <i>Seoul National University</i>	2025
BK21 Scholarship <i>National Research Foundation of Korea (NRF)</i>	2022 – 2023, 2025
Electronics Manufacturing Specialist Training Program <i>Korea Institute for Advancement of Technology (KIAT)</i>	2022 – 2023
Grand Prize, Idea Hackathon for Customs Clearance Reading Using AI <i>National IT Industry Promotion Agency (NIPA)</i>	2022

OTHER EXPERIENCE

Supporting Victims of Cyberbullying & Reducing Social Media Violence (TLAB) Collected data via Twitter and YouTube APIs; built intuitive visualizations with matplotlib, seaborn, and pyvis.	2022 – Present
Improving Medicine Access for the Visually Impaired Implemented OCR with YOLOv5 and clovaai text-recognition models to extract text from medicine labels.	2022

VOLUNTEER

Rescue Organization for Stray Cats in Redevelopment Areas	2021 – Present
Overseas Volunteer, Vietnam (Seoul National University of Science & Technology)	2020
Entrepreneurial Education Corps for Middle & High School Students	2020

REFERENCES

Prof. Hyunwoo Park

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Prof. Hakyeon Lee

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